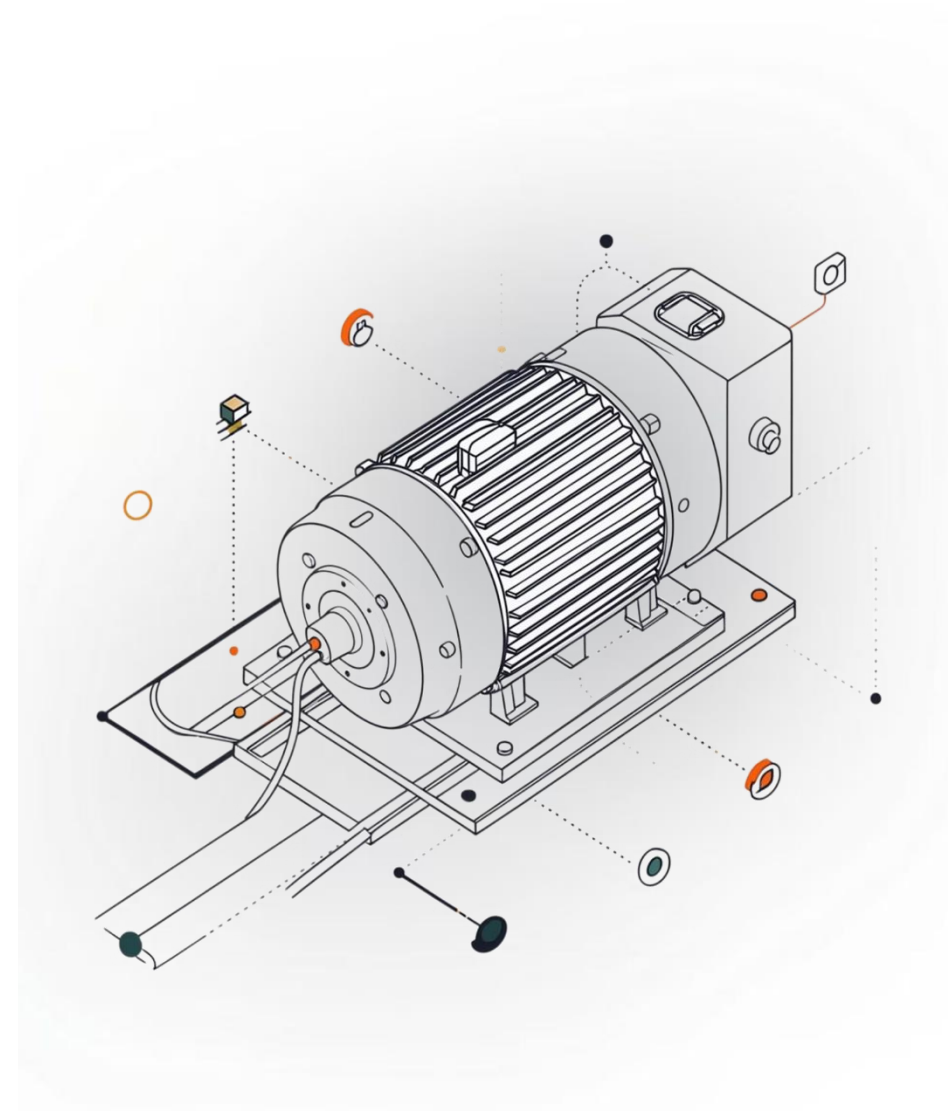


Motor Fault Detection using Motor Current Signature Analysis

Machine learning fault classification for three-phase induction motors

Ali Ashoor | General Assembly DSB PT3 | Bahrain, August 2026



The Motor is the Heart of the Plant

Motors drive every process — pumps, compressors, conveyors, fans. When the heart stops, everything stops.

MCSA is a blood test for that heart.

We read the motor's electrical current **non-invasively** and diagnose faults from its signature.

No shutdown. No disassembly.



Problem & Motivation

Unplanned Failure Cost

Downtime often costs more than the motor itself.

Traditional Detection Limits

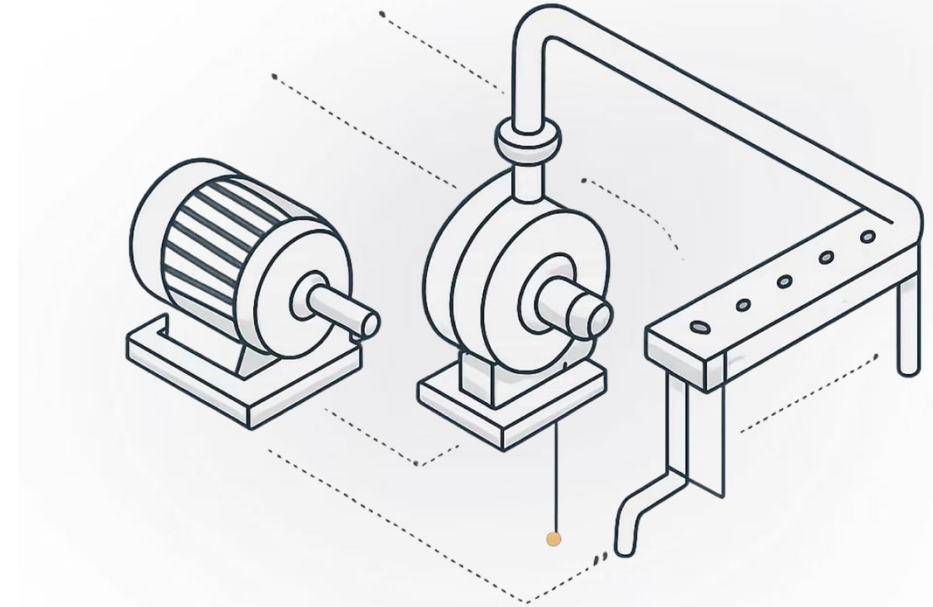
Needs vibration sensors, thermal cameras, or a full **shutdown**.

MCSA Advantage

Current is already measured by existing control equipment.

Prior System Gap

Rule-based system worked at full load, **failed** at low speed.



Aim & Objectives

Can a **machine learning classifier** match or outperform an existing **rule-based system**, and which feature extraction approach performs best?

01

Raw Signal Amplitude

Does raw amplitude carry fault information?

02

Signal Statistics

Which statistics separate **Healthy**, **BRB** and **SWF**?

03

Replicate Consistency

Are fault signatures consistent across replicates?

04

Speed Effects

Does operating speed change fault appearance?

05

Feature Extraction

Compare **TD**, **FFT**, **WT** and **hybrid** pipelines.

Dataset & Experimental Setup

Scale

2.7 million rows, 18 columns.

Conditions

Healthy, Broken Rotor Bar (BRB), and **Stator Winding Fault (SWF)**.

Operating Speeds

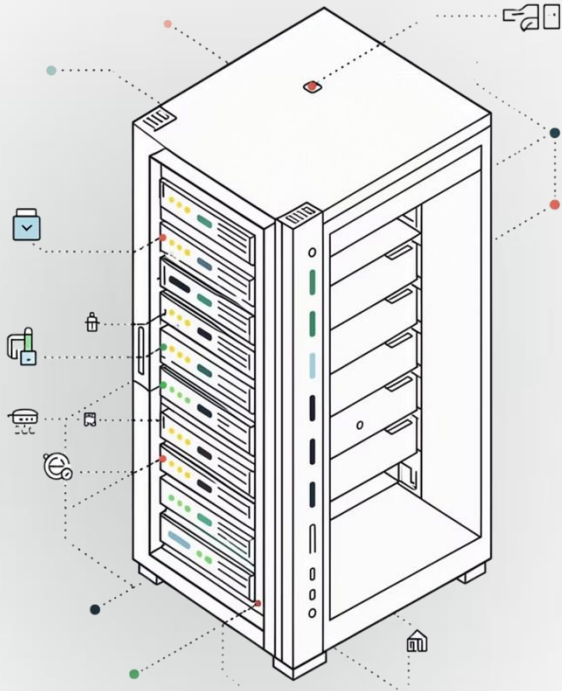
50%, 75%, and 100% of rated load.

Signal Columns

3 phases × 5 replicates = 15 signal columns, **20 kHz** sampling.

Balance

900K rows per condition — perfectly **balanced**.



Key EDA Finding

0.007

Strongest link found
between raw current and
the fault

Effectively zero.

What the data showed

Clean — No missing values, no outliers, no corrupted recordings across **2.7 million rows**.

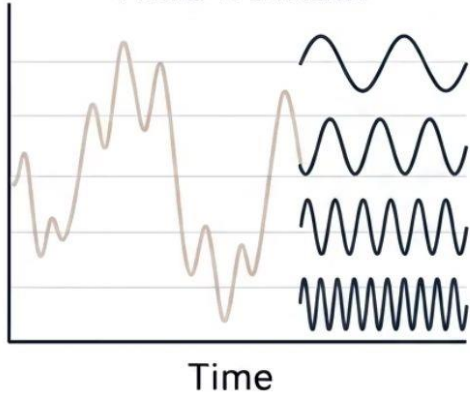
Balanced — Equal **Healthy**, **BRB** and **SWF** data at each of the three speeds.

Feature extraction is **essential** — Healthy and faulty waveforms look almost identical.

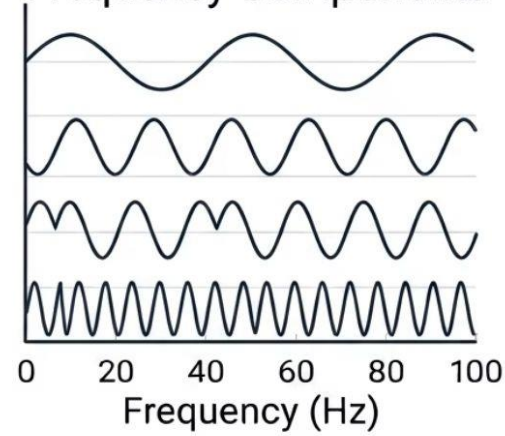
Feature extraction is not an optimization; it is what makes the problem solvable.

Scientific Data Analysis

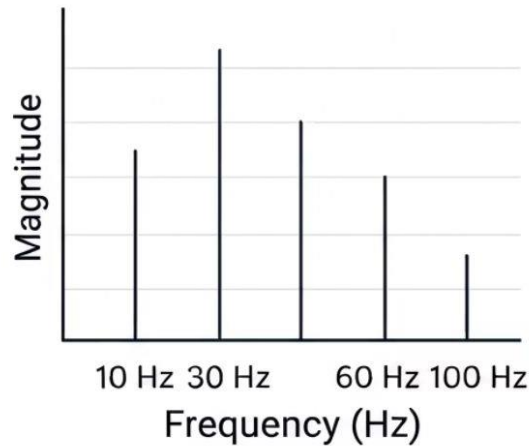
Time Domain



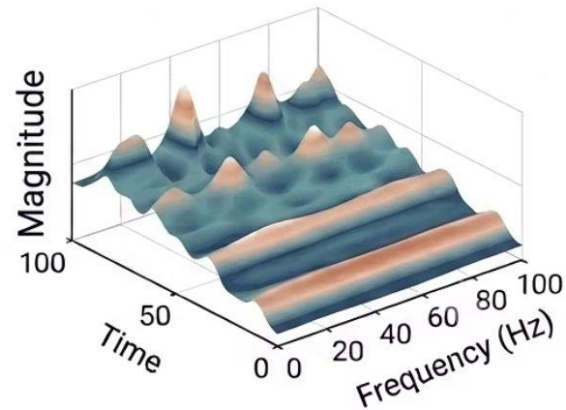
Frequency Components



Fast Fourier Transform



Wavelet Transform



How Each Pipeline Sees the Same Signal

The same **51 ms** of current, four different transformations.

01

The Raw Signal

Continuous waveform at **20 kHz**. Unreadable by a model.

02

Time-Domain (**TD**)

Shape statistics: RMS, peak, crest factor, kurtosis — **10 features**.

03

Frequency-Domain (**FFT**)

Fault harmonics at known locations — **43 features**.

04

Wavelet (**WT**)

Energy shifts over time, catching transients **FFT** misses — **8 features**.

How the Features Were Extracted

Four different ways of looking at the same 51 milliseconds of motor current.

Time-Domain — 10 features

Measures the shape of the current: how strong it is, how spiky, how steady. Values like peak, average strength, and how much it varies.

Frequency-Domain — 43 features

Breaks the current into its individual frequencies. Each fault has its own frequency fingerprint — we measure how strong the signal is at each one.

Wavelet — 8 features

Tracks how the signal changes moment to moment, not just overall. Catches short bursts that appear and disappear — which the other methods average away.

Hybrid — 61 features

All three sets combined into one dataset.

Each method asks a different question of the same signal.

ML Methodology

1

Train / Test Split

Replicates m1–m4 train, m5 test. No shared experimental run.

2

Segmentation

The recording is cut into overlapping **51 ms** slices → **63k** train / **15k** test windows.

3

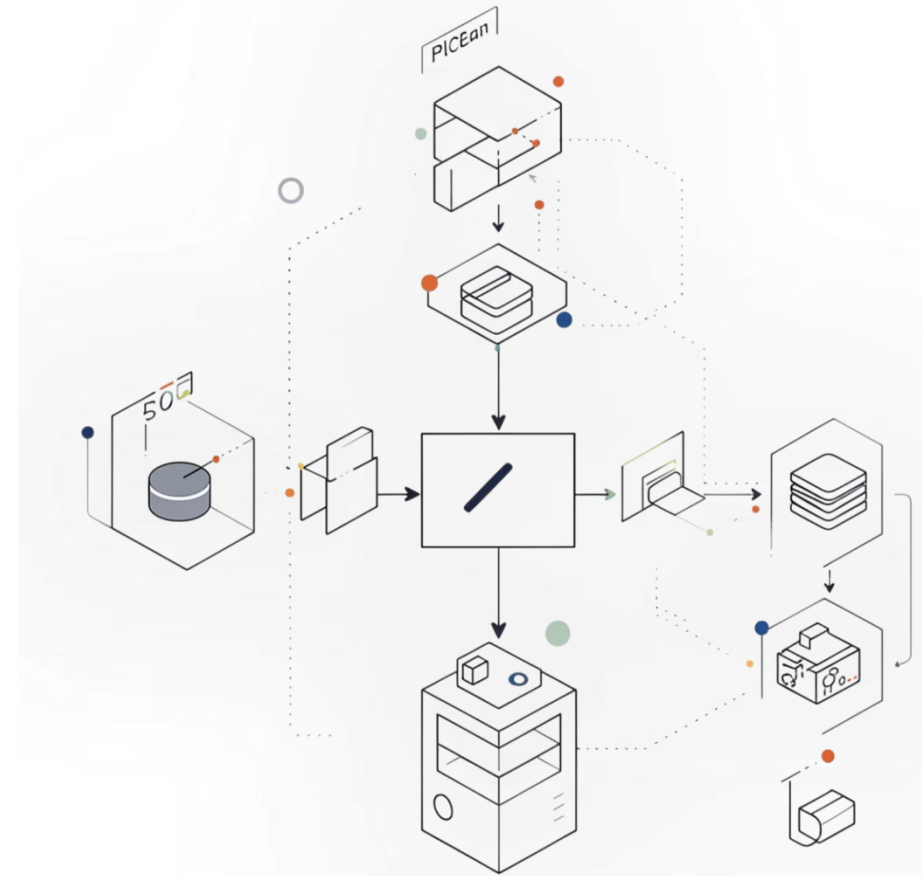
Feature Pipelines

TD (10), **FFT** (43), **CWT** (8), **HYB** (61).

4

Classifiers & Metric

Dummy, **Random Forest**, **SVM**, **XGBoost** = **16 models**. Metric: weighted F1.



16-Model Comparison

Weighted F1 — a single accuracy score from 0% to 100% — across all pipelines and classifiers. Dummy baseline = **33.71%**.

Pipeline	Dummy	Random Forest	SVM	XGBoost
TD	33.71%	83.89%	67.14%	77.11%
FFT	33.71%	86.51%	43.91%	78.04%
CWT	33.71%	84.80%	81.52%	81.90%
HYB	33.71%	98.76%	89.32%	97.50%

HYB scores highest. The next slides explain why it was not selected.

The HYB Problem

98.76%

Weighted F1 Score

The Hybrid pipeline achieved the highest score.

A score this high is a warning sign, not a win.

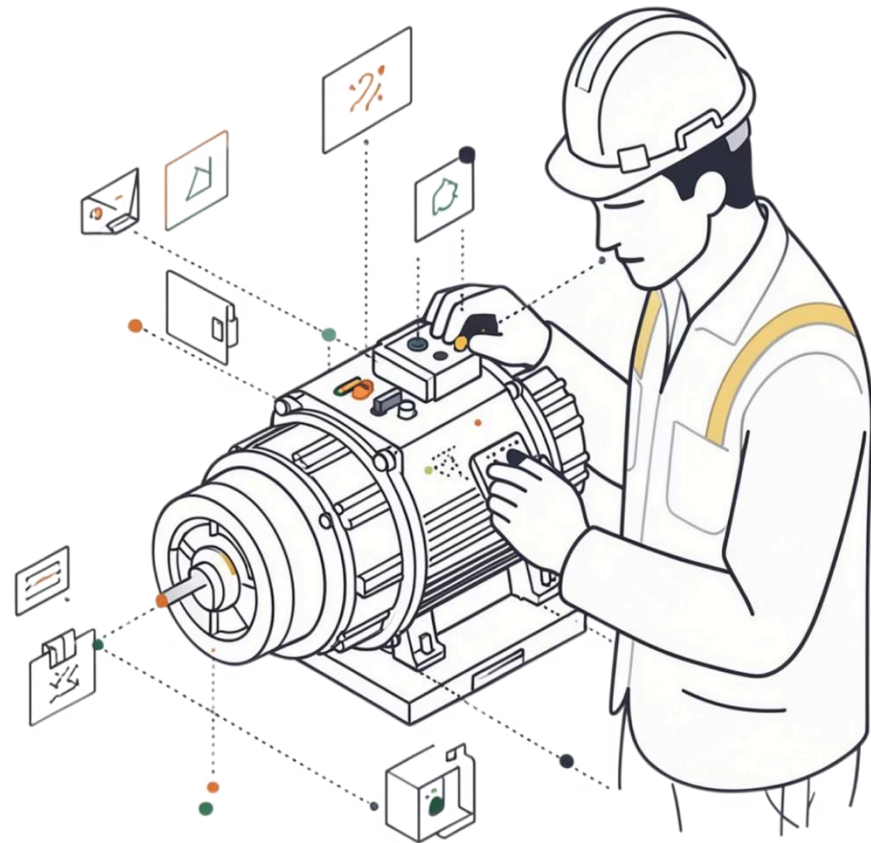
HYB combines **TD** + **FFT** + **CWT** into **61 features**.

Many of those features describe the same thing in different ways.

A model can then exploit coincidences in the numbers, not real fault physics.

The question: is it detecting faults, or just finding patterns that happen to match the labels?

A score that cannot be explained cannot be **trusted**.



Model 1: SVM on Time-Domain

The default settings were far too rigid — tuning fixed it.

Before

67.14%

Weighted F1 Score with default settings

After

85.89%

Weighted F1 Score after tuning

What it means

Gain: **+18.75 pp** — the largest improvement of the three models.

Tuning the model's settings, tested repeatedly on held-out data, made the difference.

Parameters: C = 386.5, gamma = 0.809, kernel = RBF

Model 2: Random Forest on FFT

Removing weak measurements helped more than tuning the model.

Before

86.51%

Across all **43 FFT** features.

After (Tuned)

87.16%

Only **+0.65 pp** improvement from tuning alone.

After (17 Features)

89.87%

Reduced from **43 → 17** features by removing **26** weak ones.

- ❑ Removing 26 weak features added **+2.71 pp** — over 4× the tuning gain. **Fewer, better measurements beat more measurements.**

Parameters: n_estimators = 830, max_depth = 30, max_features = None, min_samples_split = 4



Model 3: XGBoost on CWT

Even **8** measurements carried real fault information.

Before

81.90%

Across all **8 CWT** features.

After

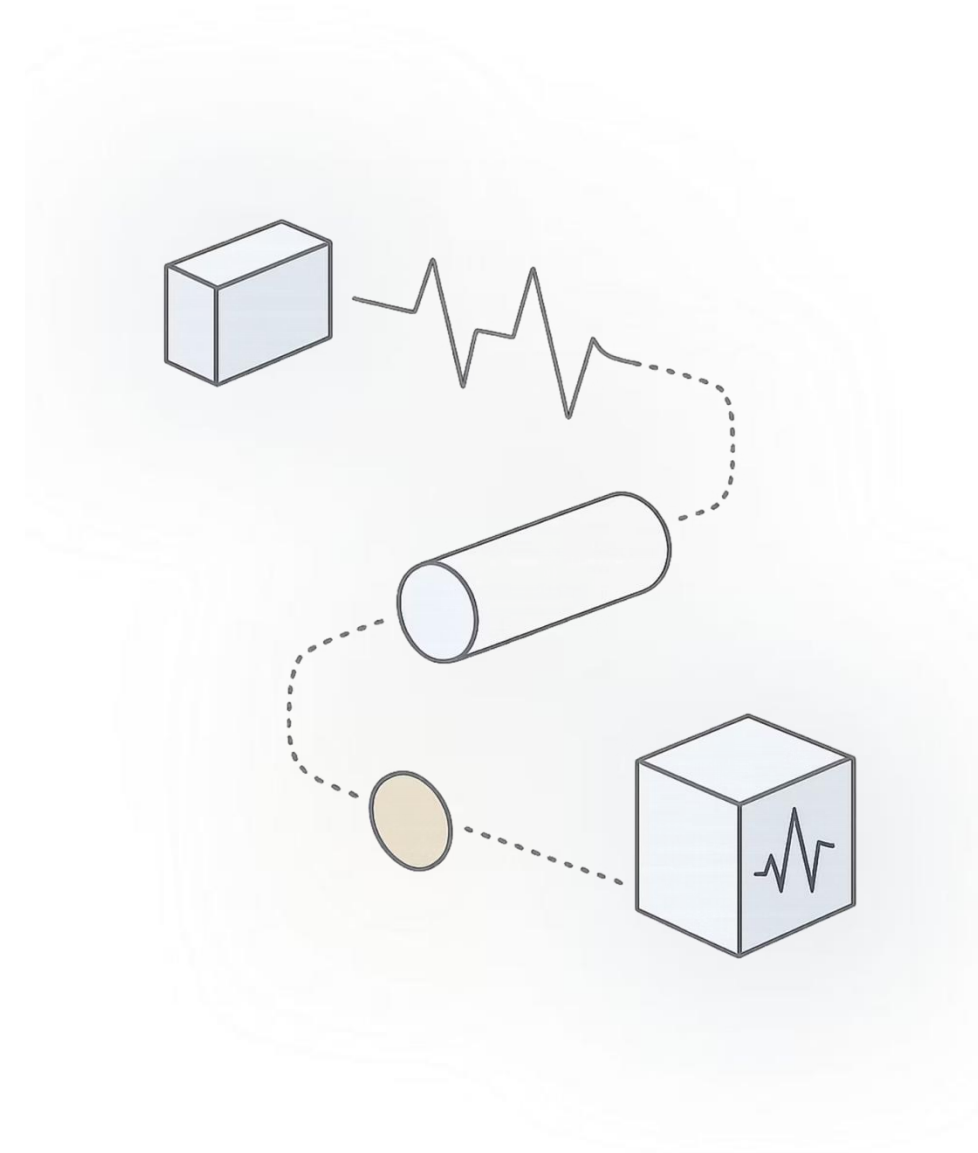
84.91%

Gain — **+3.00 pp** from tuning.

What it means

A deeper, slower-learning model was needed. The wavelet measurements carry real fault signal.

Parameters: `n_estimators = 1,087`, `max_depth = 10`, `learning_rate = 0.057`



Final Model Selection

89.87
%

Weighted F1 Score

Random Forest on **17** reduced FFT features

Per-class F1: **BRB** 85% | **Healthy** 98% | **SWF** 86%

Selected because:

Explainable — Each feature points to a specific fault frequency. An engineer can check the reasoning.

Fast — **17** measurements from a 51 ms window. Light enough to run continuously.

Trustworthy — Built on known fault signatures, not unexplained patterns.

Comparison with the Rule-Based Baseline

Rule-Based System (Prior Thesis)

Fixed thresholds calibrated for full load.

100% speed: **100%**

75% speed: **~50%**

50% speed: **~0%**

Overall: **~50%**

Machine Learning Model

89.87% weighted F1 across all three speeds simultaneously.

One classifier, no speed-specific recalibration.

The model adjusts to speed on its own.

Key Results & Engineering Interpretation



The Same Harmonic Won

The most useful feature was **H19** — the same harmonic the rule-based system used. The model learned real fault physics.



Stator vs. Rotor

16 of 17 features describe stator faults. Only one rotor feature survived — rotor faults stay the weakest class.



Healthy Motors Rarely Flagged

98% F1 for **Healthy** motors. Almost no false alarms.



Speed Generalization

One model, all three speeds. No recalibration.

Limitations

Single Motor Design

A motor built differently would need its fault frequencies recalculated.

Fixed Operating Speeds

Works only at **50%, 75%, 100%**. A new speed needs retraining.

Rotor Fault Detection

Rotor faults produce a faint signal. Reliable detection needs a longer measurement window.

Two Fault Types Only

Only **BRB** and **SWF** studied. Bearing faults, misalignment and eccentricity were not covered.

Key Takeaway

Machine learning solved what fixed rules could not.

Raw Current Tells You Nothing

Correlation of **0.007**. The signal must be split into frequencies first.

FFT + Random Forest Won

89.87% across all three speeds, in one classifier.

Fewer Features, Better Results

17 features beat **43**. Feature quality beats quantity.

The Highest Score Was Set Aside

HYB scored **98.76%** but could not be verified. What cannot be explained cannot be trusted.

☐ **One model. All three speeds. No shutdown required.**

Thank You

Questions & Discussion